

TurEngMix: A Text Corpus and Benchmark for Turkish-English Code-Mixed Language Identification and Named Entity Recognition

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Abstract

Natural language processing systems underperform on code-mixed text, particularly for low-resource language pairs. Turkish-English poses a further challenge: it lets English stems combine with Turkish suffixes to form single mixed-language tokens. We introduce TurEngMix, a corpus of 5.5K noisy, naturally occurring social media posts (486,974 tokens) rich in Turkish-English code-mixing. From this corpus, we construct a new Turkish-English benchmark for code-mixed language identification (LID) and named entity recognition (NER), comprising 15K expert-annotated tokens. Evaluating both decoder LLM and fine-tuned encoder baselines, we find that monolingual Turkish and English tokens are labeled reliably, but all models have high error rates on mixed-language tokens for both LID and NER. For morphologically integrated tokens, NER error rates were 5.2× and 6.3× higher for GPT-4o and Qwen, respectively. This highlights how morphological integration remains a challenge. We release the corpus, annotations, and code to support future computational and sociolinguistic research on Turkish-English code-mixing.

1 Introduction

Code-mixing, the use of two or more languages within a single sentence or phrase, is widespread in online spaces and diaspora communities (Poplack, 1980; Androutsopoulos, 2013). It serves a range of linguistic and social functions, including identity signaling, tone-setting, and communicating culturally or technically specific concepts, and takes different structural forms across speaker backgrounds and communicative contexts (Tay, 1989). However, despite its natural role in multilingual communities and growing presence in informal, online settings, code-mixing remains underexplored in natural language processing (NLP), particularly for lower-resourced language pairs (Winata et al.,

Token	Gloss
<i>influencerlarımız</i>	influencer -lar-ımız influencer- PL-1PL.POSS 'our influencers'
<i>frameworklerinin</i>	framework -ler-i- nin framework- PL-3PL.POSS-GEN 'of their frameworks'
<i>tweetlememesi</i>	tweet -le- me- me- si tweet- VBLZ-NEG-NMLZ-3SG.POSS 'the fact that they do not tweet'

Table 1: Mixed-language tokens with English stems and Turkish suffixes.

2023), reflecting the broader language diversity gap in NLP (Joshi et al., 2020; Pakray et al., 2025).

Existing code-mixing benchmarks are dominated by language pairs such as Spanish-English and Hindi-English (Doğruöz et al., 2021), where code-mixing typically occurs at word boundaries. Turkish, however, presents a markedly different linguistic setting. Its agglutinative morphology allows suffixes expressing case, possession, tense, and other grammatical functions to stack into complex words. In Turkish-English code-mixed discourse, English lexical stems are productively integrated into this structure (Yirmibeşoğlu and Eryiğit, 2018), resulting in *mixed-language tokens* that combine English roots with Turkish suffixes (examples in Table 1). These naturally occurring forms are linguistically interesting manifestations of bilingual language use that challenge traditional structural constraints of code-mixing (Kemaloglu-Er, 2018) and pose unique difficulties for otherwise simple foundational sequence-labeling tasks such as language identification (LID) and named entity recognition (NER).

These linguistic challenges are compounded by a scarcity of annotated resources. To our knowledge, there is only one publicly available Turkish-English code-mixed text dataset, comprising 391 naturally occurring examples with word-level lan-

guage labels (Yirmibeşoğlu and Eryiğit, 2018). We are not aware of any resources for evaluating NER for Turkish-English code-mixing, nor LID with mixed-language tokens. This paper fills that gap and makes the following contributions:

- **TurEngMix**: a large-scale corpus of naturally occurring Turkish-English code-mixed social media posts, spanning a diverse range of on-line topics. Beyond this work, it supports future computational and sociolinguistic study of lexical innovation, code-mixing behavior, and multilingual language use.
- An annotated Turkish-English benchmark with token-level language and named-entity labels (15K tokens) derived from TurEngMix.
- A systematic evaluation of decoder LLM and fine-tuned encoder baselines for LID and NER on Turkish-English code-mixed text.

We release the corpus and annotated benchmark under a CC BY-NC-SA 4.0 license for non-commercial use. The corpus, benchmark, code, and preprocessing resources are publicly available.

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2 Background

2.1 Turkish-English Code-Mixing

Code-mixing is well-documented among Turkish speakers in both diaspora communities and on-line spaces. Turkish immigrant communities in Western Europe exhibit extensive code-mixing as a norm in everyday informal in-group conversation (Backus et al., 2010), and similar patterns have been observed among Turkish-English bilinguals in the United States, where intra-sentential switching occurs at particularly high rates (Koban, 2013). Turkish-English code-mixing is also prevalent in social media contexts, where it serves a range of communicative functions including identity expression and tone-setting (Akgür, 2021).

Turkish-English written code-mixing is highly asymmetric. Turkish is agglutinative, verb-final, and suffixal. Following the Matrix Language Frame Model (Myers-Scotton, 1993), this makes it natural for Turkish to provide the matrix frame and English to supply content stems that take Turkish inflection (e.g., English nouns and adjectives bearing Turkish case, possessive, or derivational suffixes). Mixed-language tokens (i.e., English roots with Turkish suffix chains) are frequent in informal

written communication, such as in Turkish social media data (Yirmibeşoğlu and Eryiğit, 2018). Accurately processing such tokens is thus an essential prerequisite for downstream NLP modeling and analysis in this domain, but remains a challenge for extant systems.

2.2 Code-Mixing Resources and Evaluation

Most existing code-mixed NLP resources are concentrated on a small set of language pairs, mainly Spanish-English and Hindi-English (Doğruöz et al., 2021; Winata et al., 2023). Central benchmarks (e.g., LinCE (Aguilar et al., 2020) and GLUECoS (Khanuja et al., 2020)) explicitly acknowledge this skew as they provide evaluation limited to a few popular pairs. This imbalance is a core structural problem in the field (Doğruöz et al., 2023). More recent work has begun expanding corpus coverage to underrepresented language pairs. For example, VietMix (Tran et al., 2026) introduced an expert-translated Vietnamese-English corpus for machine translation, demonstrating both the utility of carefully curated resources and the continued difficulty that modern models face on code-mixed text. Nevertheless, similar resources remain unavailable for many morphologically and typologically distinct language pairs, including Turkish-English.

While performance on lower-level tasks such as LID and NER has improved for code-mixed data (Winata et al., 2023; Pakray et al., 2025; Sheth et al., 2025), these tasks remain challenging in realistic multilingual settings. Recent studies have shown that LID performance can degrade across domains (van der Goot, 2025), and code-switched LID remains difficult even across multiple language pairs, with existing approaches struggling to reliably identify language boundaries (Burchell et al., 2024). Similarly, multilingual NER benchmarks continue to expose challenges involving noisy text, fine-grained entity categories, and complex entity structures (Malmasi et al., 2022). Recent work has also explored specialized approaches for code-mixed LID, including training-free methods such as MaskLID (Kargaran et al., 2024). However, there remains a gap between higher-resourced and lower-resourced language pairs. Moreover, performance on more complex NLP tasks such as sentiment analysis and question answering still lags behind that of non-mixed data. For example, CodeMixBench, a multilingual benchmark spanning eighteen languages (though no Turkic languages), reports persistent performance degradation on code-mixed in-

¹<https://huggingface.co/datasets/ilydoa/TurEngMix>

²<https://github.com/kemnguyenle/TurEngMix>

ui ux designer and front end developer olarak uygun fiyatlı **b2c b2b**, wordpress, e-ticaret gibi çeşitli alanlarda **web siteleri** yapabilirim. ulaşmak isteyenler yeşillendirsın.
tanım : sertbest çalışan
Topic Tag: freelance

Figure 1: Example *Eksi Sözlük* post collected for the TurEngMix corpus, with code-mixing (English words mixed with Turkish text) highlighted for reference.

puts across a range of NLP tasks despite recent advances in LLMs (Yang and Chai, 2025).

There are some resources for tasks such as sentiment analysis and question-answering in Turkish (Çöltekin et al., 2023), and transformer models trained specifically for Turkish (Schmitt and Schweter, 2026; Uludoğan et al., 2024). For code-mixing, there exists a publicly available Turkish-German Twitter/X corpus (Çetinoğlu, 2016) and an audio corpus (Çetinoğlu, 2017). However, we are only aware of one Turkish-English code-switched written corpus, which comprises 391 naturally-occurring samples collected from social media settings and is designed for the task of code-switching detection (Yirmibeşoğlu and Eryiğit, 2018). To our knowledge, TurEngMix is the first Turkish-English benchmark jointly annotated for named entity recognition and token-level language identification on naturally occurring forum text, and the largest such resource to include explicit metadata for English-origin stems with Turkish morphology.

3 TurEngMix Corpus

We first introduce the data collection and filtering steps to create the TurEngMix corpus and present corpus statistics. In the next section, we describe our annotated benchmark for LID and NER, followed by our evaluation of baseline models.

3.1 Data Collection

The corpus consists of public Turkish social media posts from the Turkish-language social media site *Eksi Sözlük*, a Reddit-like social media site with posts organized by topic tags, which function similarly to subreddits. The filtered dataset, token-level annotations, and preprocessing scripts are publicly available.

We collected 30K publicly-available posts using the eksipy API wrapper. Because posts could be collected by their topic tag category, we first curated a diverse list of topic tags that were likely

Statistic	Value
Unique posts	5,549
Total tokens	486,974
Post length range	3–500 words
Mean post length	87.75 words
Median post length	57 words

Table 2: Summary statistics of the TurEngMix corpus.

to surface code-mixing, including tech topics, social media/entertainment, lifestyle, education, and more (see Appendix for the full topic list). The posts were retrieved as text strings. No usernames, profile information, timestamps, or other personally identifying metadata were retained.

3.2 Data Filtering

To increase the prevalence of Turkish-English code-mixing, we applied a three-stage filtering pipeline. First, the *langdetect* Python library was used to identify posts containing both Turkish and English. We then applied two GPT-4o filtering stages: the first performed a binary check for the presence of English lexical material, while the second repeated this while excluding posts in which apparent English content consisted solely of URLs, brand names, or other non-linguistic strings that could produce false positives. Lastly, posts containing fewer than three words or more than 500 words, where a word is defined as a whitespace-separated character sequence, were excluded after filtering.

To assess the filtering pipeline, we manually inspected 100 retained and 100 discarded posts. Of the retained posts, 71 contained Turkish-English code-mixing, compared with only 22 of the discarded posts, indicating that filtering substantially raised the density of code-mixed content relative to the unfiltered pool. Not every retained post is code-mixed, and some code-mixed posts are discarded. The retained set, therefore, is strongly enriched for code-mixing rather than exhaustive. The remaining monolingual posts serve as useful negative examples for downstream LID and NER analyses.

3.3 Corpus Statistics

The final TurEngMix Corpus consists of 5,549 unique posts comprising 486,974 tokens (Table 2). Posts have a mean length of 87.75 words and a median length of 57 words. Figure 2 shows the distribution of post lengths, illustrating that while most posts are relatively short, the corpus also captures longer-form text posts characteristic of online

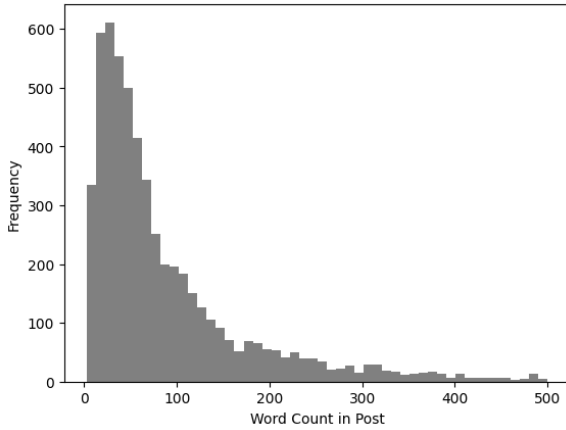


Figure 2: Word count distribution in the final dataset. Post word count was majority within 100 words, with longer posts scaling to 500.

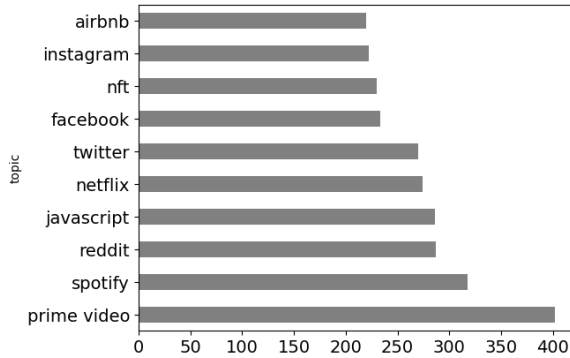


Figure 3: Ten most-frequent topic tags in the final dataset are primarily tech and entertainment.

forums.

The corpus spans 71 unique topic tags. Technology and social media related topics (e.g., *twitter*, *netflix*, and *javascript*) are among the most frequently represented categories (Figure 3). The prominence of technology and social media topics is consistent with the documented dominance of English in online and technical domains (Danet and Herring, 2007), which makes such settings especially conducive to code-mixing (Androutsopoulos, 2013; Akgür, 2021).

The corpus captures a broad spectrum of code-mixing behavior, including isolated English lexical insertions, longer mixed-language phrases, and morphologically integrated English stems with Turkish suffixes (Table 3). As the TurEngMix corpus preserves naturally occurring Turkish-English discourse across diverse online contexts, it supports future investigations of code-mixing behavior, multilingual communication, and language variation, while also providing a resource for developing and

evaluating NLP systems beyond LID and NER.

4 LID and NER Annotated Benchmark

To enable standardized evaluation of foundational sequence labeling tasks on Turkish-English code-mixing, we constructed an annotated benchmark comprising 15,012 annotated tokens from the TurEngMix Corpus. By deriving the benchmark directly from naturally occurring social media posts, the annotated dataset preserves the linguistic characteristics of authentic Turkish-English code-mixed discourse.

4.1 Benchmark Construction

We first randomly sampled 250 posts from the corpus, then tokenized each post with a custom Python script that separates text based on whitespace and punctuation boundaries while preserving alphanumeric strings and morphologically mixed tokens. We removed hyperlinks, non-linguistic symbols, and special characters that did not contribute to language identification or named entity annotation. Punctuation marks such as apostrophes or quotation marks were retained because of their role in building morphologically complex words. The tokenization script and annotation guidelines are publicly available alongside the released dataset.

4.1.1 Annotation Scheme

Each token received a language identification label and a named entity label, as shown in Figure 4.

LID We adopted the annotation framework of Solorio et al. (2014), labeling each token as TR, EN, MIXED, OTHER, AMBIGUOUS, or NE. Following the shared task convention, all named entities were assigned the NE label regardless of language of origin or the presence of Turkish suffixes. For example, *netflix'in* was labeled NE, not MIXED, despite containing a Turkish possessive suffix.

NER For NER, we followed the CALCS 2018 shared task guidelines (Aguilar et al., 2018). Named entities were annotated as Person, Organization, Location, Group, Product, Title, Event, Time, or Other using the BIO tagging scheme.

Code-Mixing Metadata We also annotated for a broader "INTEGRATED" token category which captured whether an English-origin word, including named entities and brands, had a Turkish suffix. There were 386 tokens in this category, including

Code-mixing type	Original post	English translation
Isolated English insertions	her açtığımda belki gülhan şen kaydolmuş- tur diye search edip havamı aldığım trendy oluşum.	The <i>trendy</i> platform that, every time I open it, I <i>search</i> thinking ‘maybe Gülhan Şen has signed up’ and end up being disappointed.
Mixed-language phrase	love is an excuse to get hurt diye çok mantıklı bir açıklamasını yapmış birileri	Someone gave a very reasonable explanation that “ love is an excuse to get hurt. ”
Mixed-language token	Bu link ’teki video mutlaka izlenmeli bence.	The video at this link should definitely be watched.

Table 3: Different types of code-mixing found in the dataset, including isolated English insertions, longer mixed language phrases, and mixed-language tokens.

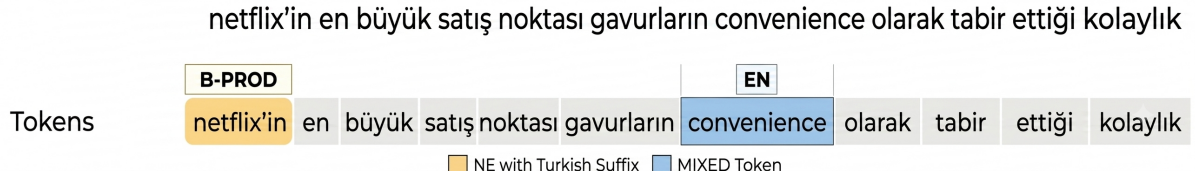


Figure 4: An example annotation sentence from the golden annotated dataset. Translation: *Netflix’s biggest selling point is the ease of use which the Westerners refer to as ‘convenience.’*

tokens that had been labeled as NE in LID annotations but contained morphological mixing. This allowed us to understand the prevalence of the tokens that displayed the morphological mixing behavior.

Posts were also labeled with the type of code-mixing present (see Table 3), including INTEGRATED tokens embedded into the post, isolated English words inserted into Turkish text, and English phrases (> 2 words) present in the text. There was overlap between categories.

4.1.2 Annotation Process

The first author, fluent in both Turkish and English, annotated each token. To validate, inter-annotator agreement was calculated with a secondary annotator (also fluent in both languages) on a sample of 203 tokens from randomly sampled posts. A Cohen’s Kappa score of 0.96 was achieved for LID and 0.94 for NER. Per-class agreement was high across categories (see Appendix A.3), with the lowest on MIXED ($\kappa = 0.75$).

Disagreements primarily involved lexicalized English loanwords (e.g., *videolar*, which may reasonably be interpreted as either Turkish or morphologically mixed) and distinctions between semantically related named entity categories (e.g., Product versus software-related terms). These cases were resolved through discussion, yielding complete agreement in the final annotations.

LID Label	Count
TR	11,723
EN	1,992
NE	1042
Mixed	189
Ambiguous	62
Other Language	4
Total	15,012

Table 4: Distribution of LID labels. Note that NE tokens included additional morphologically mixed examples.

4.2 Benchmark Statistics

The benchmark exhibits substantial code-mixing, with a mean document-level Code-Mixing Index (CMI) of 13.25 (SD = 13.12) and 87.2% of posts containing more than one language (Das and Gambäck, 2014). Although morphologically integrated tokens account for only 2.57% of all annotated tokens, they appear in 66.0% of posts, indicating that Turkish morphological integration is a frequent characteristic of code-mixed discourse despite its relatively low token frequency. Beyond morphologically integrated tokens, the benchmark also captures other common forms of Turkish-English code-mixing. A total of 154 posts contain isolated English insertions, while 148 contain longer embedded English phrases (Table 6).

For the encoder models, the benchmark was partitioned into training, validation, and test splits using an 80/10/10 ratio, with posts (rather than individual tokens) assigned to splits to prevent in-

NER Label	Count
Non-Entity (O)	13,971
Product	359
Title	246
Organization	129
Person	119
Location	88
Other	52
Group	26
Event	19
Time	3
Total	15,012

Table 5: Distribution of named entity labels.

Split	Posts
Embedded Mixed Token	165
Isolated English Token	154
Embedded English Phrase	148

Table 6: Distribution of code-mixing patterns in the benchmark.

formation leakage across sets (Table 7). Each split maintained a similar distribution of LID and NER labels, with most labels represented (See Table 4 and 5 for benchmark label distributions). For NER, some of the rarest NE labels (Event, Time, etc) were not represented in the test set.

5 Baseline Evaluations

We present baseline evaluations of pretrained encoder and decoder models on two mature sequence-labeling tasks—LID and NER.

5.1 Decoder Setup

We selected GPT-4o and Qwen3-8B to represent two complementary classes of contemporary LLMs: a large proprietary model and a compact open-weight multilingual model.

For both tasks, models received complete posts reconstructed from the benchmark as sequences of pre-tokenized, indexed tokens, rather than receiving isolated words. This preserved the original sentential context of each social media post and allowed models to leverage context when predicting on ambiguous tokens. Models were prompted to assign a label to each pre-tokenized input token while preserving its provided token index. For LID, labels corresponded to the benchmark language categories. For NER, models predicted BIO-formatted named entity labels. Predicted labels were aligned with the benchmark using the provided token indices prior to scoring, and malformed outputs were

Split	Posts	Sentences	Tokens
Train	200	254	12,466
Validation	25	26	1,173
Test	25	41	1,373
Total	250	321	15,012

Table 7: Dataset statistics across train, validation, and test splits.

treated as incorrect.

Model temperature was fixed at 0 to encourage deterministic outputs, and prompts required the model to assign exactly one label to each token. We evaluated both zero-shot and three-shot prompting to measure the effect of in-context examples. The three demonstration examples were selected to cover the range of language and named entity labels present in the benchmark. Complete prompts are included in the Appendix.

Predictions were produced for the complete benchmark (250 posts) and compared against the gold-standard annotations. We report macro-averaged F1 scores together with per-label F1 scores for both the LID and NER tasks.

5.2 Encoder Setup

To compare general-purpose LLMs with specialized encoder architectures, we evaluated three pretrained transformer encoders: XLM-RoBERTa (Conneau et al., 2020), BERTurk (Yildirim, 2024), and TurkishBERTweet (Najafi and Varol, 2023). These models represent different sources of linguistic specialization: multilingual representation learning, general Turkish language modeling, and social-media-specific Turkish language modeling.

Models were fine-tuned for token classification using the Hugging Face Transformers framework on the TurEngMix Benchmark training split, with a majority-class baseline reported for comparison. Because encoder tokenizers operate on subword units, benchmark tokens were further segmented using each model’s tokenizer, with labels assigned only to the first subword piece and continuation subwords ignored during training and evaluation.

Fine-tuning was performed using the Hugging Face Trainer with a learning rate of 2×10^{-5} , weight decay of 0.01, and a maximum of 10 epochs. Each model was trained using four random seeds. For each run, the checkpoint with the highest validation F1 score was evaluated on the held-out test split, and we report the mean and standard deviation across runs. Predictions were evaluated

against the gold-standard annotations using macro-averaged and per-label F1 scores.

6 Results

6.1 Language Identification

Distinguishing morphologically integrated English borrowings from ordinary Turkish or English tokens is the central difficulty in Turkish-English LID. Overall macro-F1 is modest across all models (Table 9): GPT-4o is strongest, with three-shot prompting giving a small gain over zero-shot (0.5170→0.5238), while Qwen3-8B does not benefit from demonstrations (0.4920→0.4788). Fine-tuned encoders are comparable overall, with BERTurk and TurkishBERTweet leading among them. The aggregate scores, however, mask sharp per-class variation (Table 8): every model labels Turkish and English tokens well ($F1 > 0.84$), but MIXED tokens and named entities are far harder. GPT-4o reaches only 0.45 F1 on MIXED tokens; the best encoder (TurkishBERTweet) reaches 0.22. Encoders match the LLMs on named entities ($F1$ 0.56–0.68) but the results are much lower on MIXED, confirming that morphological integration—not language identity per se—is what current models struggle with (Table 8).

6.2 Named Entity Recognition

Table 10 reports overall BIO-level NER performance. GPT-4o outperformed Qwen3-8B under both prompting settings, and three-shot prompting improved GPT-4o from a macro F1 of 0.5669 to 0.6140. In contrast, encoder models achieved considerably lower macro F1 despite high token-level accuracy, reflecting the strong class imbalance in the dataset where non-entity tokens account for approximately 93% of all tokens.

Because BIO evaluation jointly measures entity detection, entity classification, and boundary prediction, we additionally evaluate binary entity detection by collapsing all entity categories into entity (NE) and non-entity (Non-NE) (Table 11). GPT-4o achieved an F1 of 0.80 on the NE class under three-shot prompting, suggesting that many BIO-level errors were associated with correctly identifying the specific entity type or boundary prediction rather than failure to detect named entities. Qwen achieved a lower F1 of 0.70 on the NE class, while encoder models ranged from 0.47 to 0.67.

Per-category evaluation reveals wide variation across entity types. Person, Location, and Title

entities generally had higher F1 scores, whereas Organization, Other, Event and Time were substantially lower (Appendix Tables A3-A4). Across all evaluated models, Organization and Product were the most frequently confused categories, particularly for technology-related entities whose interpretation depended on context (e.g., *Google*, *Netflix*, and *Microsoft*).

6.3 Error Analysis

Qualitative inspection demonstrates consistent error patterns across both sequence labeling tasks.

6.3.1 LID

For language identification, both GPT-4o and Qwen frequently misclassified morphologically integrated English borrowings as ordinary Turkish tokens, particularly when multiple Turkish suffixes were attached (e.g., *entryden*, *influencerlarımız*, and *editleyip*). Named entities containing Turkish suffixes (e.g., *Microsoft’un* and *Sony’nin*) were also commonly predicted as MIXED rather than NE, suggesting that models relied heavily on surface morphology instead of treating named entities as a separate language category. Conversely, English compounds containing hyphens (e.g., *k-food* and *marketing-mix*) were often incorrectly labeled as MIXED despite containing no Turkish morphology.

6.3.2 NER

For named entity recognition, the most common errors involved morphologically integrated English borrowings and technology-related named entities. Technology entities frequently produced Product-Organization confusion (e.g., *Twitter*, *Google*, *Cloudflare*, and *Last.fm*), while morphologically integrated entities such as *reactlerle* and *python’daki* were often missed after Turkish suffix attachment.

6.3.3 Morphological Integration in Named Entities

To further examine whether morphological integration affects NER performance, we conducted an error analysis on the GPT-4o three-shot and Qwen three-shot settings. We compared NER error rates between tokens containing English-origin stems bearing Turkish suffixes (including morphologically adapted named entities) and all other evaluated tokens (Table 12). GPT-4o produced errors on 19.7% of mixed tokens compared with only 3.8% of other tokens. A chi-square test confirmed that this difference was highly significant ($\chi^2 = 231.83$, $p < 0.001$), corresponding to an

Model	Setting	TR	EN	MIXED	NE	OTHER	AMBIGUOUS
GPT-4o	zero-shot	0.93	0.89	0.42	0.62	0.19	0.00
GPT-4o	three-shot	0.98	0.90	0.45	0.67	0.13	0.00
Qwen3-8B	zero-shot	0.98	0.86	0.33	0.60	0.07	0.12
Qwen3-8B	three-shot	0.98	0.86	0.39	0.54	0.10	0.00
XLm-RoBERTa	fine-tuned	0.98	0.87	0.00	0.68	0.00	0.00
BERTurk	fine-tuned	0.97	0.84	0.13	0.64	0.00	0.00
TurkishBERTweet	fine-tuned	0.98	0.84	0.22	0.56	0.00	0.00

Table 8: Per-class F1 scores for LID. Decoder LLMs were tested on the full dataset. Encoder models were fine-tuned on the train/dev and evaluated on the test set.

Model	Setting	Macro F1
GPT-4o	Zero-shot	0.5170
GPT-4o	Three-shot	0.5238
Qwen3-8B	Zero-shot	0.4920
Qwen3-8B	Three-shot	0.4788
Baseline	–	0.1414
XLm-RoBERTa	Fine-tuned	0.4358 $\pm$ 0.0137
BERTurk	Fine-tuned	0.4734 $\pm$ 0.0355
TurkishBERTweet	Fine-tuned	0.4195 $\pm$ 0.0177

Table 9: Macro F1 scores for LID task. Decoder LLMs were tested on the full dataset. Encoders were fine-tuned on the train/dev and evaluated on the test set.

Model	Setting	Macro F1
GPT-4o	Zero-shot	0.5669
GPT-4o	Three-shot	0.6140
Qwen3-8B	Zero-shot	0.3993
Qwen3-8B	Three-shot	0.4162
Baseline	–	0.0502
XLm-RoBERTa	Fine-tuned	0.0772 $\pm$ 0.0246
BERTurk	Fine-tuned	0.1759 $\pm$ 0.0550
TurkishBERTweet	Fine-tuned	0.0821 $\pm$ 0.0273

Table 10: BIO-level NER macro F1.

odds ratio of 6.21. Qwen exhibited an even larger disparity, with error rates of 33.5% and 5.3%, respectively ($\chi^2 = 507.17$, $p < 0.001$; odds ratio = 8.84). These results suggest that morphological integration introduces an additional challenge beyond general code-mixed language processing.

7 Conclusion

This work addresses gaps in Turkish-English code-mixed NLP by introducing a large-scale corpus, an expert-annotated benchmark for language identification and named entity recognition, and systematic evaluations of current LLM and encoder-based models. Our analysis reveals that morphologically mixed tokens remain a persistent challenge for multilingual NLP systems, highlighting an underexplored difficulty for agglutinative languages

Model	Setting	Non-NE	NE
GPT-4o	Zero-shot	0.98	0.78
GPT-4o	Three-shot	0.98	0.80
Qwen3-8B	Zero-shot	0.97	0.70
Qwen3-8B	Three-shot	0.98	0.73
XLm-RoBERTa	Fine-tuned	0.97	0.67
BERTurk	Fine-tuned	0.96	0.47
TurkishBERTweet	Fine-tuned	0.97	0.62

Table 11: F1 scores for binary NER. BIO labels are collapsed into two categories: NE and NON-NE.

Model	Other Tokens	Mixed Tokens
GPT-4o	3.8%	19.7%
Qwen3-8B	5.3%	33.5%

Table 12: Comparison of NER error rates between morphologically integrated English borrowings and other evaluated tokens.

in code-mixed settings. These findings suggest that progress on code-mixed NLP for agglutinative languages may require morphology-aware approaches, such as sub-token language attribution or morphological segmentation as preprocessing, rather than scale alone. By releasing these resources and evaluations, we aim to support future research on multilingual language understanding and improve the representation of diverse language varieties in NLP.

8 Limitations

As the data was collected from *Ekşi Sözlük*, a social media site with a specific user base, this Turkish-English code-mixing resource is thus only most accurately representative of the code-mixing practiced by that online community, and does not account for different patterns in other domains. Furthermore, because topic tags likely to contain Turkish-English code-mixing were prioritized during corpus construction, the corpus is not intended to estimate the prevalence of code-mixing in Turkish online communication. As the site is public,

pretraining contamination is possible. Future work could expand to other online communities or other settings (e.g. spoken, in-person).

A further consideration is that GPT-4o served both as a filtering component during corpus construction (§3.2) and as an evaluated baseline. Posts in the corpus are, by construction, posts in which GPT-4o detected English lexical material, which may favor GPT-4o’s LID performance relative to models that played no role in dataset creation. We partially mitigate this by including a langdetect-based first stage and by manual inspection of retained and discarded posts, but results for GPT-4o should be interpreted with this selection effect.

The annotation scheme follows established task definitions, but the separate NE category can obscure cases where named entities also exhibit Turkish morphological integration; we partially address this by separately annotating the "INTEGRATED" category for all English-origin tokens with Turkish morphological suffixes. The LLM prompts also did not systematically provide demonstrations for every annotation category, leaving the effect of alternative prompting strategies as an important direction for future work.

Another limitation was the class imbalance present in the dataset. Some LID and NER categories contained only a small number of instances, limiting the reliability of per-category evaluation for rare labels. For LID, the majority of tokens were TR tokens. For NER, O (non-entity) tokens comprised the majority of the dataset.

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A Appendix

A.1 Dataset Collection

The TurEngMix Corpus was collected from publicly available posts on *Ekşi Sözlük* during November 2025. Candidate posts were retrieved using topic tags selected to increase the likelihood of Turkish-English code-mixing.

A.1.1 Topic Tags

```
topic_names = [  
    #Tech / Programming / AI  
    "spotify", "google", "python", "ai",  
    "chatgpt", "openai", "midjourney", "dalle",  
    "javascript", "react", "nodejs", "git",  
    "blockchain", "web3", "crypto", "nft",  
    "startup", "venture capital",  
  
    #Social Media / Internet Culture  
    "facebook", "instagram", "twitter",  
    "youtube", "tiktok", "linkedin", "discord",  
    "reddit",  
    "influencer", "meme", "shitpost", "gaming",  
    "twitch", "streamer",  
  
    #Entertainment / Gaming / Anime  
    "netflix", "prime video", "disney+",  
    "minecraft", "lol", "anime", "kpop",  
    "manga", "webtoon",  
    "rap", "hiphop", "edm", "dj",  
  
    #Sports  
    "futbol", "basketbol", "tennis", "formula  
    1", "voleybol", "gym", "workout", "fitness",  
  
    #Lifestyle / Travel / Food / Shopping  
    "yemek", "tatil", "seyahat", "alışveriş",  
    "fashion", "sneakers", "nike", "adidas",  
    "apple", "samsung", "mobile", "app",  
    "airbnb", "hotel", "vegan", "fitness  
    lifestyle",  
  
    #Daily Life / Work / Education  
    "üniversite", "iş", "freelance",  
    "marketing", "growth", "seo", "branding",  
    "startup life",  
    "gündem", "aşk", "sağlık", "covid",  
    "ekonomi", "politika", "haberler",  
    "finance",  
  
    #Misc / Fun / Pop Culture  
    "lol memes", "viral", "challenge", "trend",  
    "gaming memes", "internet slang",  
    "funny videos", "edutainment", "tech news",  
    "apps review", "music", "concert",  
    "festival"  
]
```

A.2 Corpus Filtering

Corpus construction employed a multi-stage filtering pipeline to identify posts containing Turkish-English code-mixing while removing noise.

A.2.1 Stage 1 Prompt: English Word Detection

Initial filtering prompt used to identify posts containing English words.

```
prompt_text = """
Detect if the given Turkish social media post
contains any English words. If English
words are present, return true.

Steps
1. Language Detection: Detect whether the given
text contains English words.

Output Format:
Detection Result: Output "True" if English
words are present, otherwise "False".
Number your answers to match the post
numbers, and put each response on its own
line.

Example output:
1. True
2. False
etc
"""
```

A.2.2 Stage 2 Prompt: Code-Mixing Verification

Second-stage filtering prompt providing a more explicit instructions for detection of Turkish-English code-mixing.

```
prompt_text = """
You will be given Turkish social media posts.
Go through the entire post, word by word,
and detect if it contains any English words
mixed into the Turkish, including
individual English words or
sentences/phrases in English. Don't count
URLs or brand names as English words.

If English words are present in the post,
return true.

Steps
1. Language Detection: Detect whether the given
post contains English words.

Output Format:
Detection Result: Output "True" if English
words are present, otherwise "False".
Number your answers to match the post numbers,
and put each response on its own line.

Example output:
1. True
2. False
etc
"""
```

A.2.3 Language Identification Prompt

Prompt template used for LLM-based language identification experiments.

```
prompt_text = """
Determine the language label for each token in
the input text.

Use the following labels:
TR: Turkish
EN: English
MIXED: Mixed Turkish and English
OTHER: Any other language
NE: Named entity (representing all or part of a
named entity)
AMBIGUOUS: Could belong to either language in
context

The input consists of numbered tokens. Each
input line contains an integer index,
followed by a tab character, followed by a
token.

For every input line:
- Output exactly one line.
- Preserve the same INDEX.
- Assign exactly one language label.
- Do not skip any indices.
- Do not duplicate any indices.
- Do not renumber the indices.

Output each line in the following format:

INDEX|LABEL

The number of output lines must exactly equal
the number of input lines.

Return only the INDEX|LABEL pairs, one per
line. Do not include the original tokens,
explanations, numbering, headings, blank
lines, or any other text.
"""
```

A.2.4 Named Entity Recognition Prompt

Prompt template used for LLM-based named entity recognition experiments.

```
prompt_text = """
Analyze the following text and identify named
entities using the BIO tagging scheme.

Use the following BIO format:
- B-: The first token of a named entity.
- I-: Any subsequent token belonging to the
same named entity.
- O: A token that is not part of any named
entity.

Use the following entity categories:
- B-PER / I-PER (Person): Proper names or
nicknames that uniquely identify a real or
fictional person, including artists and
famous people.
- B-ORG / I-ORG (Organization): Companies,
institutions, corporations, and other
entities that have members or employees and
act as a whole. When an entity could refer
to either an organization or another
category (e.g., Facebook), use the
```

surrounding context to determine the correct label.

- B-LOC / I-LOC (Location): Physical places that people can visit, including countries, cities, addresses, facilities, landmarks, restaurants, and tourist attractions. When an organization name is used to refer to a physical place, label it as LOC.
- B-GROUP / I-GROUP (Group): Sports teams, music bands, duos, and similar groups of people.
- B-PROD / I-PROD (Product): Manufactured products, medicines, food products, well-defined services, website applications, website accounts, devices, and similar products. Distinguish products from organizations using context.
- B-TITLE / I-TITLE (Title): Titles of creative works such as movies, books, TV shows, songs, and similar media.
- B-EVENT / I-EVENT (Event): Named events such as concerts, competitions, conferences, festivals, and award ceremonies. Do not use this category for holidays.
- B-TIME / I-TIME (Time): Months, weekdays, seasons, holidays, and recurring calendar dates. Do not label hours, minutes, seconds, or relative temporal expressions such as "yesterday," "tomorrow," "week," or "year."
- B-OTHER / I-OTHER (Other): Named entities that do not fit the categories above, including nationalities, languages, music genres, and similar entities.
- O (None): Not part of a named entity

The input consists of numbered tokens. Each input line contains an integer index, followed by a tab character, followed by a token.

For every input line:

- Output exactly one line.
- Preserve the same INDEX.
- Assign exactly one named entity label.
- Do not skip any indices.
- Do not duplicate any indices.
- Do not renumber the indices.

Output each line in the following format:

INDEX|LABEL

The separator between INDEX and LABEL must be the vertical bar character "|".

Do not use tabs.

Do not use spaces.

Correct:

1|B-PER

Incorrect:

1 B-PER

1 B-PER

1: B-PER

The number of output lines must exactly equal the number of input lines.

Return only the INDEX|LABEL pairs, one per line. Do not include the original tokens, explanations, numbering, headings, blank

lines, or any other text.
"""

A.3 Per-Class Inter-Annotator Agreement

Table A1 displays inter-annotator agreement for LID per class. Table A2 displays inter-annotator agreement for NER per class. All disagreements were resolved.

LID Label	Cohen's κ
TR	0.98
EN	0.98
NE	0.94
MIXED	0.75

Table A1: Per-class inter-annotator agreement for language identification (LID).

NER Label	Cohen's κ
B-LOC	1.00
I-LOC	1.00
B-PER	1.00
I-PER	1.00
B-PROD	1.00
B-TITLE	1.00
I-TITLE	1.00
O	0.94
B-OTHER	0.00

Table A2: Per-class inter-annotator agreement for named entity recognition (NER). The low κ score for B-OTHER is likely due to the small number of instances in this category.

A.4 Additional Experimental Results

A.4.1 Per-Class Named Entity Recognition

Table A3 presents per-class named entity recognition results for GPT4o and Qwen.

Table A4 presents per-class named entity recognition for the encoder models.

A.5 Error Analysis Examples

Table A5 presents representative model errors involving Turkish morphological integration and named entity ambiguity.

A.6 Released Resources

The following resources are publicly available:

- TurEngMix Corpus
- TurEngMix Benchmark
- Annotation guidelines

Model	O	ORG	PER	LOC	PROD	GROUP	TITLE	EVENT	TIME	OTHER
GPT-4o Zero-shot	0.98	0.48	0.79	0.83	0.53	0.35	0.69	0.46	0.20	0.29
GPT-4o Few-shot	0.98	0.57	0.83	0.84	0.67	0.42	0.72	0.43	0.09	0.34
Qwen3-8B Zero-shot	0.97	0.43	0.55	0.65	0.46	0.35	0.66	0.00	0.09	0.43
Qwen3-8B Few-shot	0.98	0.47	0.63	0.64	0.52	0.28	0.53	0.00	0.07	0.34

Table A3: Per-class F1 scores for decoder LLMs on NER task. O denotes non-entity tokens.

Model	O	LOC	ORG	PER	PROD	TITLE	OTHER
Baseline	1.00	0.00	0.00	0.00	0.00	0.00	0.00
XLM-RoBERTa	0.97	0.00	0.00	0.00	0.28	0.00	0.00
BERTurk	0.97	0.00	0.091	0.44	0.27	0.14	0.00
TurkishBERTweet	0.96	0.00	0.00	0.00	0.27	0.06	0.00

Table A4: Per-class F1 scores for named entity recognition using encoder models. O denotes non-entity tokens.

- Dataset construction scripts
- Training and evaluation scripts
- LLM prompting scripts
- Fine-tuning code

Token	Gold Label	Model Prediction	Error Explanation
<i>influencerlar</i>	MIXED	TR	English-origin stem <i>influencer</i> with Turkish plural/possessive morphology. The token requires identifying both the English lexical root and Turkish morphological integration.
<i>google'a</i>	B-PROD	B-ORG	In context, <i>Google</i> refers to the search product/service rather than the organization.
<i>sony'nin</i>	NE	MIXED	The token contains a Turkish possessive suffix attached to a named entity. The model incorrectly prioritized morphological mixing over named entity identification.

Table A5: Representative error cases involving morphological integration and named entity ambiguity.